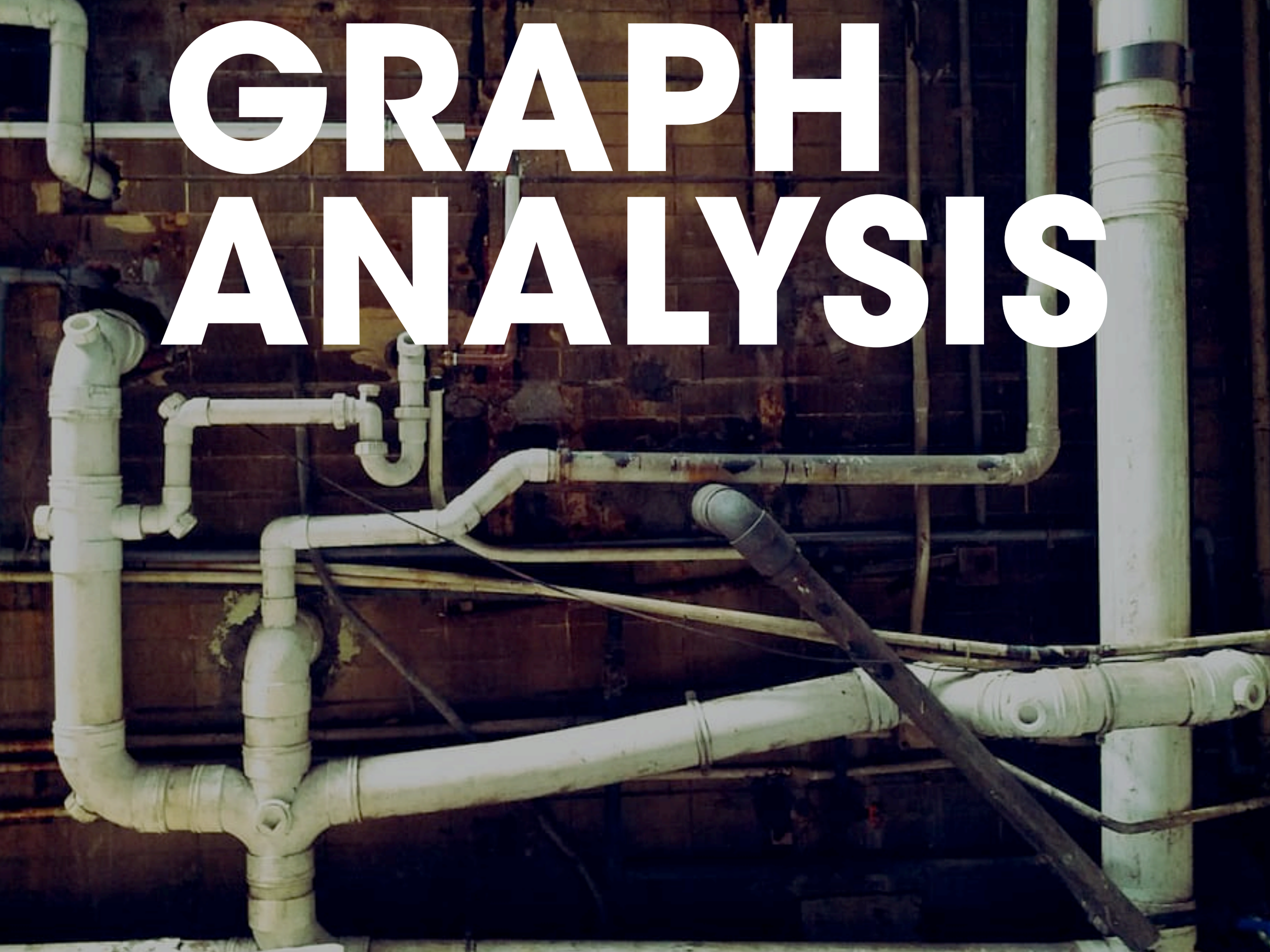GRAPH ANALYSIS